Requirement Analysis Phase

Solution Requirements(Functional & Non Functional)

Date	28-10-2026
Team ID	SWTID-2026-7875
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Executive Summary & System Purpose

This document details the functional and non-functional requirements for deploying a script-controlled
Access Control List (ACL) solution in ServiceNow. The solution enforces granular database-level access rules
across custom records in the u_institution_details table. Access rights are derived dynamically by evaluating
user role memberships alongside field-level record attributes (specifically, department branch alignment).

2.Functional Requirements (FR):

- FR-1: Identity & Role Provisioning (`EEE User`, roles `bb1`–`bb4`)
- FR-2: Data Schema Configuration (`u_institution_details` table and attributes)
- FR-3: Row-Level Read ACL Filtering (`Branch == EEE` script evaluation)
- FR-4: Granular CRUD Rights (Create: `bb2`, Write: `bb3`, Delete: `bb4`)

Req ID	Requirement Name	Detailed Specifications & Implementation Scope
FR-1	Identity & Role Provisioning	• Provisioned EEE User (eeeuser@gmail.com) • Created custom operational roles: bb1 (Read), bb2 (Create), bb3 (Write), bb4 (Delete) • Assigned all four roles to the EEE User account
FR-2	Data Schema Configuration	• Configured u_institution_details custom table • Fields: Student Roll Number (Auto-Number), Student Name & Faculty Name (sys_user Ref) • Branch Choice List (EEE, ECE, CSE), Email, Phone Number, and Multi-line Description
FR-3	Row-Level Read ACL Filtering	• Executed Read ACL on u_institution_details requiring role bb1 • Script condition: Evaluates Branch == 'EEE' for standard branch users • Preserved unrestricted access for System Administrator (gs.hasRole('admin'))
FR-4	Granular CRUD Rights	• Create ACL: Assigned to role bb2 (controls visibility of 'New' button) • Write ACL: Assigned to role bb3 (enforces field modification rights) • Delete ACL: Assigned to role bb4 (restricts record deletion in list/form view)

3.Non-Functional Requirements (NFR):

- NFR-1: Latency & Performance Optimization (<15ms evaluation overhead)
- NFR-2: Database-Level Security Enforcement & direct URL protection
- NFR-3: Maintainability & Admin Override Fail-safes

Req ID	Requirement Name	Performance, Security & Governance Criteria
NFR-1	Latency & Performance	• Script evaluation overhead executed in under 15 milliseconds per query cycle • List view rendering for u_institution_details completed within 1.5 seconds • Zero impact on overall ServiceNow database processing speeds
NFR-2	Database Security Enforcement	• Access policies enforced at database engine level (not UI layer only) • Prevents unauthorized access via direct URL endpoints (e.g., /u_institution_details.do) • Mandatory security_admin role elevation required for security modifications
NFR-3	Maintainability & Admin Fail-safes	• Explicit admin override logic included prior to standard rule processing • Prevents administrative lockout during script condition failures • Adheres to ECMAScript 2021 (ES12) standard JavaScript guidelines